

Haroune Mohammedi | Senior Data Engineer

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Hello there! I'm Haroune, a Senior Data Engineer with a passion for building scalable data solutions. With over 5 years of experience in the field, I have a proven track record of designing and implementing data pipelines, optimizing database performance, and leveraging big data technologies to drive business insights. I thrive in collaborative environments and am always eager to learn and adapt to new challenges in the ever-evolving world of data engineering.

EXPERIENCE

ENGIE | SENIOR DATA ENGINEER

Feb 2024 – Present | Paris, France

- Designed and implemented large-scale data pipelines processing energy consumption data at 30-minute frequency, handling high-volume datasets in a Big Data environment (Databricks, PySpark).
- Migrated legacy ETL workflows from PL/SQL to PySpark on Databricks, reducing execution time from several weeks to a few hours and generating substantial compute cost savings.
- Optimized distributed data processing through Spark performance tuning, partitioning strategies, and scalable architecture design.

QUADRATIC | SENIOR DATA ENGINEER

May 2022 – Jan 2024 | Paris, France

- Design and implement the backend of a Smart Contract App with Python.
- Design the Mongodb database schema, the GraphQL schema and queries.
- Setup the integration tests environment, write unit/integration tests.

BIGMAMA TECHNOLOGY | SENIOR DATA ENGINEER

July 2019 – April 2022 | Algiers, Algeria

- Core developer of our multi-cloud MLOps platform <https://iko.ai> to train, track, package, deploy, and monitor machine learning models.
 - Automatically tracking machine learning experiments.
 - One-click model packaging (Docker) and deployment (Kubernetes).
- Analyze requirements, design and build software to solve the problem.
- Compare open-source solutions, synthesize, and integrate the best solution.

BIGMAMA TECHNOLOGY | DATA ENGINEER

September 2017 – June 2019 | Algiers, Algeria

- Built and deployed a data-intensive application using Akka, Spark, Kafka, Elasticsearch, and Firebase, focusing on high-availability and fault tolerance.
- Developed an internal SDK using Scala for integrating different dynamic parts of our stack (Firebase, Kafka, Akka ..ect), focusing on maintainability and extensibility.
- Contributed on the design of many complex applications based on the actor model.
- Provisioned Ansible recipes for internal information system setup/security (Gitlab, Apt Mirror, SFTP, 2FA Authentication).

UNIVERSITY AND PERSONAL PROJECTS

HOMELAB | PROXMOX, ANSIBLE, PYTHON

2025 - Present | Personal Project

- Designed and built a personal home automation system using Proxmox, Ansible, and Python.

WEB APPLICATION FOR NLP | PYTHON

2019 | University Project

- Frontend: A modern web interface using Vuejs and Vuetify.
- Backend: A RESTful API using Django and Django Rest Framework (DRF).
- Based on microservice architecture, dockerized and deployed on GKE.

VIDEO PLAYER CONTROLLED BY HAND GESTURES | SCALA, AKKA

2018 | University Project

- Asynchronous and reactive JavaFx based app.
- Actor model architecture using Akka and Akka remote.
- Hand gesture detection neural network model trained with Keras.

SKILLS

Proficient: Python • PySpark • Databricks

Experienced: Scala/Akka • HTML/CSS/JS • SQL

Familiar: Java • Shell • Vuejs • Kafka • Spark

MLOps: MLFlow • SelonCore • Jupyter • Istio • Envoy

Tools/Platforms/OS: Linux • Latex • Git • Docker • Kubernetes • GCP

EDUCATION

UNIVERSITY OF HOUARI BOUMEDIEN BAB EZZOUAR

PhD in Artificial Intelligence

September 2019 - Present | Algeria
Data services management in multi-cloud environments

Master in Artificial Intelligence

September 2017 - June 2019 | Algeria
Final project: Graph-based recommender system.

Bachelor in Computer Science

September 2014 - June 2017 | Algeria
Final project: Web services relationship prediction using Spark in collaboration with University of Michigan, USA.

REFERENCES

Pr. Youcef Aklouf, Professor at USTHB | R&D Manager at Renault, France

 yaklouf@yahoo.fr

Jugurtha Hadjar, CTO, BigMama Technology, Algeria

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